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Ecuación en derivadas parciales

Definición 1 (Ecuación en derivadas parciales). Una ecuación en derivadas parciales (EDP) es una ecuación funcional de la forma:

$$F(x_1, x_2, \dots, x_n, u_{x_1}, u_{x_2}, \dots, u_{x_n}, u_{x_1 x_2}, \dots) = 0 \tag{1}$$

donde $x = (x_1, x_2, \dots, x_n) \in \Omega \subset \mathbb{R}^n$ abierto y $u\colon \Omega \longrightarrow \mathbb{R}$ es la función incógnita.
Las variables $u_{x_{i_1} \cdots x_{i_k}}$ denotan $\dfrac{\partial^k u}{\partial x_{i_1} \cdots \partial x_{i_k}}$ y $F\colon \Omega \times \mathbb{R} \times \mathbb{R}^n \times \cdots \longrightarrow \mathbb{R}$ es una función dada con un número finito de variables.

Referenciado en

- `edp-casi-lineal`
- `sol-clasica-edp`
- `ecu-ondas-dim1`
- `edp-casi-lineal-o1-sistema-caracteristico`
- `ecu-calor-dim1`
- `edp-lineal-homogenea`
- `edp-lineal`
- `orden-edp`

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